Boitumelo Lamola, 229246346

Nsovo Maswanganyi, 224788213

Nacia Sibiya, 229620205

20/30

Assignment 01 Draft

Due Date: 20 July 2026

Process of ID

[17]

1.

UX

What is the current user experience (UX)

The current user experience is very poor for everyone involved. Subsistence farmers often produce surplus crops. However, they cannot sell or donate their excess produce because they live in rural areas with poor transport infrastructure. As a result, the surplus produce goes to waste causing farmers to feel guilty and lose potential income. [At the same time, local traders want affordable fresh produce but find it difficult to identify nearby farmers who have stock. Food insecurity exists in many households even when there is plenty of food available nearby. This disconnect creates an inefficient and frustrating experience for everyone involved.]

UX other than frustrating?

−3

Would all users/stakeholders have the same UX?

Usability

What usability goals are currently met or not met

None of the usability goals are met because there is no system that connects farmers, traders, transport providers and households in need. The system is **ineffective** because users cannot easily sell, buy, donate or transport surplus produce. It is also **inefficient** as people have to rely on verbal communication to get information to the right ears. **Utility** is also not met because farmers cannot advertise their surplus produce, traders cannot easily search for nearby suppliers, transport providers cannot view delivery requests and households cannot find available food. Furthermore, users experience low **satisfaction** because food is wasted while some families remain hungry

Why is this inefficient?

−1

Stakeholders/users

Who are the stakeholders and specific users of the proposed system

The primary users of the proposed system include small scale farmers, traders, vulnerable households and transport providers. Small scale farmers would use the system to list their surplus produce. Traders such as markets, hawkers and spaza shop owners would purchase the fresh produce listed on the system. Households could use the system to access donated or reasonably priced food. Transport providers would help to help to transport the produce from farmers to traders or households.

Boitumelo Lamola, 229246346

Nsovo Maswanganyi, 224788213

Nacia Sibiya, 229620205

Assignment 01 Draft

Due Date: 20 July 2026

System developers are also important stakeholders because they are responsible for designing, developing, testing and maintaining the platform to ensure that it meets users' needs and functions reliably. Other stakeholders include non-governmental organisations (NGOs) who aim to support small farmers and improve food insecurity.

Good

How and why would your proposed solution address the identified challenge (you could introduce the type of solution you have in mind here)

Solution

The solution to address the challenge of surplus crops, poor transport infrastructure and food insecurity in many households is to introduce the radio broadcast system. Steps to introduce the system starts through community meetings that are led by local leaders to inform the rural farmers about the new radio system that will allow them to announce their surplus crops and also explain to them how it works and its purpose, how the farmers can call or send messages to the radio station.

Then collaborate with a community radio station that already broadcasts the rural areas to introduce a slot that can be called a farmers marketing time which can be scheduled at a specific time daily so that farmers and traders know when to tune in.
The farmers can use their phones to call in the station directly or SMS and to the farmers who do not have phones can use local leaders as intermediaries who are going to collect information and relay it to the station. The farmers can then announce information about their products and quantity. Traders and transport providers can respond by contacting the farmers directly or through intermediaries like local leaders to request the products and keep the broadcast relevant.

This would imply that some "users" might not be direct users as they would be dependent on others for information.
-2

The reason we propose the radio broadcast as the solution is to make it more accessible to rural farmers in the Eastern Cape as they may not be exposed to advanced technology like smartphones or online systems. Radio is familiar and it is used the most by rural areas, transport providers in their cars tune in every time to listen, most households and spaza shops have it. They don't need to learn new tools, they just call to announce their surplus produce on radio channels like Vukani community Radio. These Radio broadcasts will make the surplus visible to transporters or households. Radio also saves costs as it requires minimal investment. Farmers know when to announce and households know when to tune in which build trust and assure them that the system is reliable. This solution will relieve all the frustration, and it will empower farmers to share their surplus, earn income and reduce wastes while households get to access food affordably.

Boitumelo Lamola, 229246346

Nsovo Maswanganyi, 224788213

Nacia Sibiya, 229620205

Assignment 01 Draft

Due Date: 20 July 2026

Which two design principles would be most appropriate to consider for your solution

Not convinced that "visibility" is addressed here. −2

Visibility: The radio broadcast system is affordable and widely used, all rural farmers are exposed to it, and anyone can access it, it is inclusive to all the stakeholders, and they can use it regardless of their technological background.

Consistency: Radio broadcast is what they are familiar with, they don't need to learn about new tools. Farmers and traders should know that announcements occur regularly at a specific time so they can rely on the system which builds confidence among farmers and households. It ensures that the radio broadcast system becomes a part of daily life.

2.

(3)

a) Lesson three: What it really takes to scale Design Solutions, primarily focusing on the principle of "**Designing for Implementers**".

b) A synopsis from the principle of "Designing for implementers" stipulates that people who will be operating the solution that is devised hold critical knowledge that the designers may not know about. In this way, they must be treated as end-users of the system as well, instead of being consulted afterwards.

In this scenario, I would identify everyone who will implement the system, such as farmers, hawkers, spaza shop owners, transporters, and vulnerable households.

How would you ensure that trust is built between all involved parties? −2

Rather than merely asking these stakeholders what they think is necessary for the system to meet their needs, I would spend time observing and analysing their everyday routines. This would reveal realities that already exist, such as trusted networks, time constraints and other informal systems already in place. This would prove to be more effective since the stakeholders may not mention all the necessary components if asked.

By centering implementers early, I ensure that the eventual design properly fits into the current networks and how stakeholders are operating, instead of forcing them to adopt a new design that might break and collapse after being introduced to real-world logistics, relationships, and constraints. **[184]**
